



VNiVERSiDAD
D SALAMANCA

Elaboración de libros electrónicos mediante código y asistentes de Inteligencia Artificial

Curso para las Facultades de Ciencias y de Ciencias Químicas

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Ceci est un exemple de livre interactif créé avec la philosophie “Tout comme code”.

Utilisez le menu de gauche pour naviguer à travers les chapitres.

- Mathématiques
 - *Équations du second degré*
- Informatique
 - *Exercices Semaine 3 : Programmation III*
- Recherche
 - *Système multi-agent pour l'évaluation cervicale*
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Part I

Mathématiques

ÉQUATIONS DU SECOND DEGRÉ

Une équation du second degré est une équation polynomiale de la forme: $ax^2 + bx + c = 0$ (eq-second-degre)

Où x représente la variable, et a , b et c sont des constantes avec $a \neq 0$.

1.1 Formule quadratique

La solution à l'équation eq-second-degre est donnée par la formule quadratique: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Cette formule permet de trouver les racines de n'importe quelle équation du second degré.

1.2 Exemple de code

Le code suivant en Python calcule les racines d'une équation du second degré en utilisant la bibliothèque standard `cmath` pour gérer d'éventuelles racines complexes:

Part II

Informatique

EXERCICES SEMAINE 3 : PROGRAMMATION III

CONTENU : Classes Wrapper et premiers exercices de POO

2.1 Exercices obligatoires :

1. Vous devez créer une classe `Person.java` avec les attributs `nom`, `taille` et `poids`, ainsi que leurs getters et setters correspondants. Dans la méthode principale, vous devez instancier 3 objets de cette classe et compléter leurs informations en les demandant à l'utilisateur via la console pour chaque objet. Enfin, vous devez afficher dans la console qui est le plus grand et qui pèse le plus. Vous devez toujours utiliser les getters et setters pour obtenir et définir les informations.
2. Améliorez l'exercice 1 pour qu'il existe un constructeur sans paramètres qui initialise les valeurs de `nom`, `taille` et `poids` à des valeurs par défaut que vous jugez appropriées.
3. Améliorez l'exercice 1 pour que la classe `Person` propose une méthode calculant l'IMC de la personne à travers les valeurs actuelles des attributs de l'objet (son état). Il faut envisager la possibilité que les valeurs n'aient pas été initialisées avec des valeurs valides.
4. Analysez le flux d'exécution du programme en plaçant des points d'arrêt dans le constructeur et dans les différentes parties du code de l'exercice 1. Définissez des valeurs aux attributs directement lors de leur déclaration et vérifiez ce qui s'exécute en premier : le constructeur ou la déclaration de l'attribut.

2.2 Exercices d'extension :

5. En utilisant les méthodes `parseXXX` des classes wrapper, construisez une classe dotée de méthodes adéquates pour lire tous les types de données numériques (`int`, `float`, `double`) sous forme de chaînes de caractères. Si l'utilisateur n'insère pas de valeur valide, demandez à nouveau.
6. Modifiez l'exercice 1 pour que la classe `Person.java` se trouve dans un package appelé `model` au lieu du même package que la classe `App.java`. Qu'avez-vous dû changer pour l'utiliser dans `App.java` ?
7. Quelle est la différence entre le constructeur par défaut d'une classe en Java et un constructeur sans paramètres ?
8. Pourquoi utiliser des getters et des setters au lieu d'accéder directement aux attributs via le point ? Encapsulation. Est-ce une question de sécurité ou de conception ?
9. Quelle est la différence entre définir un attribut comme `public`, `private` ou sans modificateur de visibilité ? Dans quel cas est-il possible d'y accéder via le point (.) avec une référence d'objet ?
10. À quoi sert le mot-clé `this` ? Et le mot-clé `new` ?
11. Il est possible de définir des classes qui ont à leur tour des attributs d'autres classes (appelé Composition). Vous devez maintenant modifier l'exercice 1 pour que la classe `Person` ait un attribut de type `DNI` (une autre classe que vous devrez créer) qui aura comme attributs `numéro` et `lettre`. Au constructeur de la classe `Person`, vous devrez passer en paramètre un numéro de DNI et un objet de type `DNI` sera créé, générant la lettre correspondante à partir de l'algorithme. Consultez l'opérateur `%` en Java.

12. Créez une classe appelée `Rectangulo.java` avec les attributs `longueur` et `largeur`, chacun avec une valeur par défaut de 1. Elle doit avoir des méthodes pour calculer le périmètre et l'aire du rectangle, ainsi que des getters et setters. Les setters doivent vérifier que la longueur et la largeur sont des nombres à virgule flottante supérieurs à 0.0 et inférieurs à 20.0. Écrivez un programme pour tester la classe `Rectangulo`.
13. Créez un nouveau projet avec la classe `Person.java`. Réalisez les instructions suivantes et répondez aux questions posées.

```
public static void main(String[] args) {  
    // Création d'un objet de la classe Person  
    Person paco = new Person("Paco", 22, 177.3f, 82.3f);  
  
    // Nous copions la référence de l'objet précédent vers une autre référence  
    Person refPaco2 = paco;  
  
    // Nous utilisons la deuxième référence pour définir une valeur  
    refPaco2.setAgeYears(77);  
  
    // Résultat ?  
    System.out.printf("L'âge de %s est:%d%n", paco.getName(), paco.getAgeYears());  
  
    // Quel est l'âge affiché ?  
    // Combien d'objets y a-t-il en mémoire ? Combien de références ? Vérifiez le_  
    ↪ débogueur  
}
```

Part III

Recherche

SYSTÈME MULTI-AGENT POUR L'ÉVALUATION CERVICALE

Ce chapitre est basé sur l'article "A novel multiagent system for cervical motor control evaluation and individualized therapy: integrating gamification and portable solutions" [MBR+24].

3.1 Résumé de l'étude

L'étude s'est concentrée sur la conception d'un dispositif portable et objectif pour évaluer et traiter les troubles du Contrôle Moteur Cervical (CMC). Ce dispositif est basé sur une architecture proposée qui utilise une technologie avancée pour évaluer et améliorer le CMC des patients.

Lors d'une étude pilote avec 10 participants, la faisabilité et la convivialité du dispositif ont été vérifiées, y compris une évaluation initiale utilisant le *Head Relocation Test* et une intervention de 12 sessions sur 4 semaines.

L'architecture du système proposé est chargée de recueillir des données pertinentes sur le contrôle moteur cervical des patients. Elle utilise des algorithmes avancés pour traiter ces données et évaluer objectivement la fonction CMC. De plus, le système adapte la thérapie aux besoins individuels de chaque patient.

Les résultats préliminaires indiquent que le dispositif et l'architecture proposée ont un impact positif sur la précision des tests d'évaluation. Bien que des tests de validation supplémentaires soient nécessaires, ce dispositif apparaît comme une alternative prometteuse et précieuse pour évaluer et traiter les patients atteints de troubles CMC. Son accent sur les technologies avancées et l'adaptation personnalisée s'aligne sur les recherches précédentes en télé-réadaptation.

3.2 Architecture du système

Le système utilise un environnement multi-agent qui coordonne le flux d'informations et personnalise les exercices thérapeutiques. À cette fin, il intègre des capteurs miniatures (*Inertial Measurement Units*, IMU) qui collectent des métriques de mouvement en temps réel, corrigeant les écarts via des algorithmes de fusion de données.

3.3 Références de l'article

Les références complètes de l'étude sont : [AbdollahiMKuberPMShiraishiMSoangraRRashediE, AiQLiuZMeng-WLiuQXieSQ, BarraOrtizHAMatamalaAMInostrozaFLArayaCLMondacaVN, BarzegarKhanghahAFernieGRFekrA, BisioIGaribottoCLavagettoFSciarroneA, BlasHSSMendesASEncinasFGSilvaLAGonzalezGV, CalvaresiDMarinoniMDragoniAFHilfikerRSchumacherM, ChenJOrCKChenT, ChenKBSestoMEPontoKLeonardJMasonAVanderheidenGWilliamsJRadwinRG, DePauwJMercelisRHallemanAMichielsSTruijenSCrasPDHertoghW, DonSDKoon-ingMVoogtLickmansKDaenenLNijjsJ, DuQBaiHZhuZ, GrazioliETranchitaEBorrielloGCerulliCMingantiCParisiA, GuidanceM, HidalgoPerezAFGarciaALdUraldeVillanuevaIGMartinezAPAlemanA FernandezCarneroJLToucheR,

HumayunMJhanjhiNZAlmotilagAAlmufarehMF, KazeminasabSNejadghaderiSAAmiriPPourfathiHAKhodaeiMSullmanMJKolahiAASafiriS, KheraPKumarN, KimWSChoSKuJKimYLeeKHwangHJPaikNJ, KumarSRavellaUKShrivastavaAKAnilMSwamySK, LaranjoLDunnAGTongHLKocaballiABChenJBashirRSurianDGallegoBMagrabiFLauAYeal, MalfietAKregelJCoppietersIPauwRDMeeusMRousselNCagnieBDanneelsLNijsJ, MichielsSHertoghWDTruijenSNovemberDWuytsFHeyningPV, MihajlovicZPopovicSBrkicKCosicK, ParkJSJungYJLeeG, PengBYangLLiYLiutLiYa, PengBYangLLiYLiutLiYb, QuNTianHDMartinoEZhangB, QuNTianHDMartinoEDZhangB, RiffittsMOhAAle-muAPatelVSmithCNMuratiSBailesAAllenMDombrowskiMLeeJYeal, RodaCRodriguezACLJaqueroVNavarroEGonzalezP, SpencerJWolfSLKesarTM, SremakaewMJullGTreleavenJUthaikhupS, SulisEMarianiSMontagnaS, TreleavenJa, TreleavenJb, VinoskiS, WeizmanYTiroshOFussFKTanAMRutze, YoonTLKimHNMinJH, ZoeteRMJOsmotherlyP-GRivettDASnodgrassSJ]

3.4 Bibliographie de cette page

Part IV

Références

BIBLIOGRAPHIE

- [MBR+24] André Filipe Sales Mendes, Héctor Sánchez San Blas, Fátima Pérez Robledo, Juan F. De Paz Santana, and Gabriel Villarrubia González. A novel multiagent system for cervical motor control evaluation and individualized therapy: integrating gamification and portable solutions. *Multimedia Systems*, 2024. doi:10.1007/s00530-024-01328-6.
- [AbdollahiMKuberPMShiraishiMSoangraRRashediE] Abdollahi, M., Kuber, P.M., Shiraishi, M., Soangra, R., Rashedi, E. Kinematic analysis of 360 turning in stroke survivors using wearable motion sensors. *sensors* 22 (1), 385 (2022).
- [AiQLiuZMengWLiuxQXieSQ] Ai, Q., Liu, Z., Meng, W., Liu, Q., Xie, S.Q. Machine learning in robot assisted upper limb rehabilitation: a focused review. *IEEE Transactions on Cognitive and Developmental Systems* (2021).
- [BarraOrtizHAMatamalaAMInostrozaFLArayaCLMondacaVN] Barra Ortiz, H.A., Matamala, A.M., Inostroza, F.L., Araya, C.L., Mondaca, V.N. Efficacy of biofeedback in rehabilitation of musculoskeletal disorders: a systematic review. *postepy rehabilitacji* 36 (1), 41 (2022).
- [BarzegarKhanghahAFernieGRFekrA] Barzegar Khanghah, A., Fernie, G., Roshan Fekr, A. Design and validation of vision-based exercise biofeedback for tele-rehabilitation. *sensors* 23 (3), 1206 (2023).
- [BisioIGaribottoCLavagettoFSciarroneA] Bisio, I., Garibotto, C., Lavagetto, F., Sciarrone, A. When ehealth meets iot: a smart wireless system for post-stroke home rehabilitation. *IEEE Wirel. Commun.* 26 (6), 24–29 (2019) subjects with whiplash-associated disorders or non-specific neck pain and healthy controls. *clin. biomech.* 22 (8), 865–873 (2007). <https://doi.org/10.1016/j.clinbiomech.2007.05.008>.
- [BlasHSSMendesASEncinasFGSilvaLAGonzalezGV] Blas, H.S.S., Mendes, A.S., Encinas, F.G., Silva, L.A., Gonzalez, G.V. A multiagent system for data fusion techniques applied to the internet of things enabling physical rehabilitation monitoring. *appl. sci.* 11 (1), 331 (2020).
- [CalvaresiDMarinoniMDragoniAFHilfikerRSchumacherM] Calvaresi, D., Marinoni, M., Dragoni, A.F., Hilfiker, R., Schumacher, M. Realtime multi-agent systems for telerehabilitation scenarios. *artif. intell. med.* 96, 217–231 (2019).
- [ChenJOrCKChenT] Chen, J., Or, C.K., Chen, T. Effectiveness of using virtual reality– supported exercise therapy for upper extremity motor rehabilitation in patients with stroke: systematic review and meta-analysis of randomized controlled trials. *J. med. internet res.* 24 (6), 24111 (2022).
- [ChenKBSestoMEPontoKLeonardJMasonAVanderheidenGWilliamsJRadwinRG] Chen, K.B., Sesto, M.E., Ponto, K., Leonard, J., Mason, A., Vanderheiden, G., Williams, J., Radwin, R.G. Use of virtual reality feedback for patients with chronic neck pain and kinesiophobia. *IEEE Trans. Neural Syst. Rehabil. Eng.* 25 (8), 1240–1248 (2016).
- [DePauwJMerzelisRHallemanSAMichielsSTruijensCrasPDHertoghW] De Pauw, J., Merzelis, R., Halleman, A., Michiels, S., Truijens, S., Cras, P., De Hertogh, W. Cervical sensorimotor control in idiopathic cervical dystonia: a cross-sectional study. *brain behav.* 7 (9), 00735 (2017). <https://doi.org/10.1002/brb3.735>.
- [DonSDKooningMVoogtLickmansKDaenenLNijsJ] Don, S., De Kooning, M., Voogt, L., Ickmans, K., Daenen, L., Nijs, J. The effect of visual feedback of the neck during movement in people with chronic whiplash-associated disorders: an experimental study. *J. orthop. sports phys. ther.* 47 (3), 190–199 (2017). <https://doi.org/10.2519/jospt.2017.6891> 10.2519/jospt.2017.6891 131 page 18 of 18 a. f. sales mendes et al.

- [DuQBaiHZhuZ] Du, Q., Bai, H., Zhu, Z. Intelligent evaluation method of human cervical vertebra rehabilitation based on computer vision. *sensors* 23 (8), 3825 (2023).
- [GrazioliETranchitaEBorrielloGCerulliCMingantiCParisiA] Grazioli, E., Tranchita, E., Borriello, G., Cerulli, C., Minganti, C., Parisi, A. The effects of concurrent resistance and aerobic exercise training on functional status in patients with multiple sclerosis. *curr. sports med. reports* 18 , 452–457 (2019). [https:// doi. org/ 10. 1249/ jsr. 00000 00000 000661](https://doi.org/10.1249/jsr.0000000000000661).
- [GuidanceM] Guidance, M. Motion guidance. [https:// www. motio nguid ance. com/](https://www.motionguidance.com/). accessed 31 agu 2023.
- [HidalgoPerezAFGarciaALdUraldeVillanuevaIGMartinezAPAlemanyAFernandezCarneroJLToucheR] Hidalgo-Peréz, A., Fernández-García, Á., López-de-UraldeVillanueva, I., Gil-Martínez, A., Paris-Alemany, A., FernándezCarnero, J., La Touche, R. Effectiveness of a motor control therapeutic exercise program combined with motor imagery on the sensorimotor function of the cervical spine: a randomized controlled trial. *int. j. sports phys. ther.* 10 (6), 877–892 (2015).
- [HumayunMJhanjhiNZAlmotilagAAlmufarehMF] Humayun, M., Jhanjhi, N.Z., Almotilag, A., Almufareh, M.F. Agent-based medical health monitoring system. *sensors* 22 (8), 2820 (2022).
- [KazeminasabSNejadghaderiSAAmiriPPourfathiHAKhodaeiMSullmanMJKolahiAASafiriS] Kazeminasab, S., Nejadghaderi, S.A., Amiri, P., Pourfathi, H., Araj-Khodaei, M., Sullman, M.J., Kolahi, A.-A., Safiri, S. Neck pain: global epidemiology, trends and risk factors. *bmc musculoskelet. disord.* 23 (1), 1–13 (2022).
- [KheraPKumarN] Khera, P., Kumar, N. Role of machine learning in gait analysis: a review. *j. med. eng. technol.* 44 (8), 441–467 (2020).
- [KimWSChoSKuJKimYLeeKHwangHJPaikNJ] Kim, W.-S., Cho, S., Ku, J., Kim, Y., Lee, K., Hwang, H.-J., Paik, N.-J. Clinical application of virtual reality for upper limb motor rehabilitation in stroke: review of technologies and clinical evidence. *j. clin. med.* 9 (10), 3369 (2020).
- [KumarSRavellaUKShrivastavaAKAnilMSwamySK] Kumar, S., Ravello, U.K., Shrivastava, A.K., Anil, M., Swamy, S.K. A review on human cervical fatigue measurement technologies and data analysis methods.
- [LaranjoLDunnAGTongHLKocaballiABChenJBashirRSurianDGallegoBMagrabiFLauAYeal] Laranjo, L., Dunn, A.G., Tong, H.L., Kocaballi, A.B., Chen, J., Bashir, R., Surian, D., Gallego, B., Magrabi, F., Lau, A.Y., et al. Conversational agents in healthcare: a systematic review. *j. am. med. inform. assoc.* 25 (9), 1248–1258 (2018).
- [MalflietAKregelJCoppietersIPauwRDMeeusMRousselNCagnieBDanneelsLNijsJ] Malfliet, A., Kregel, J., Coppieters, I., Pauw, R.D., Meeus, M., Roussel, N., Cagnie, B., Danneels, L., Nijs, J. Effect of pain neuroscience education combined with cognition-targeted motor control training on chronic spinal pain. *jama neurol.* 75 (7), 808 (2018). [https:// doi. org/ 10. 1001/ jaman eurol. 2018. 0492](https://doi.org/10.1001/jaman.neurol.2018.0492).
- [MichielsSHertoghWDTruijensSNovemberDWuytsFHeyningPV] Michiels, S., Hertogh, W.D., Truijens, S., November, D., Wuyts, F., Heyning, P.V. The assessment of cervical sensory motor control: a systematic review focusing on measuring methods and their clinimetric characteristics. *gait posture* 38 (1), 1–7 (2013). [https:// doi. org/ 10. 1016/ j. gaitp ost. 2012. 10. 007](https://doi.org/10.1016/j.gaitpost.2012.10.007).
- [MihajlovicZPopovicSBrkicKCosicK] Mihajlovic, Z., Popovic, S., Brkic, K., Cosic, K. A system for head-neck rehabilitation exercises based on serious gaming and virtual reality. *multimed. tools appl.* 77 , 19113–19137 (2018).
- [ParkJSJungYJLeeG] Park, J.-S., Jung, Y.-J., Lee, G. Virtual reality-based cognitive– motor rehabilitation in older adults with mild cognitive impairment: a randomized controlled study on motivation and cognitive function. *healthcare* 8 , 335 (2020).
- [PengBYangLLiYLiutLiuYa] Peng, B., Yang, L., Li, Y., Liu, T., Liu, Y. Cervical proprioception impairment in neck pain-pathophysiology, clinical evaluation, and management: a narrative review. *pain ther.* 10 (1), 143–164 (2021). [https:// doi. org/ 10. 1007/ s40122- 020- 00230- z](https://doi.org/10.1007/s40122-020-00230-z).
- [PengBYangLLiYLiutLiuYb] Peng, B., Yang, L., Li, Y., Liu, T., Liu, Y. Cervical proprioception impairment in neck pain-pathophysiology, clinical evaluation, and management: a narrative review. *pain ther.* 10 , 143–164 (2021).
- [QuNTianHDMartinoEZhangB] Qu, N., Tian, H., De Martino, E., Zhang, B. Neck pain: do we know enough about the sensorimotor control system? *front. comput. neurosci.* (2022). [https:// doi. org/ 10. 3389/ fncom. 2022. 946514](https://doi.org/10.3389/fncom.2022.946514).

- [QuNTianHMartinoEDZhangB] Qu, N., Tian, H., Martino, E.D., Zhang, B. Neck pain: do we know enough about the sensorimotor control system? *front. comput. neurosci.* (2022). <https://doi.org/10.3389/fncom.2022.946514>.
- [RiffittsMOhAAlemuAPatelVSmithCNMuratiSBailesAAllenMDombrowskiMLeeJYea] Riffitts, M., Oh, A., Alemu, A., Patel, V., Smith, C.N., Murati, S., Bailes, A., Allen, M., Dombrowski, M., Lee, J.Y., et al. Functional range of motion of the cervical spine in cervical fusion patients during activities of daily living. *j. biomech.* 152, 111528 (2023).
- [RodaCRodriguezACLJaqueroVNavarroEGonzalezP] Roda, C., Rodríguez, A.C., López-Jaquero, V., Navarro, E., González, P. A multi-agent system for acquired brain injury rehabilitation in ambient intelligence environments. *neurocomputing* 231, 11–18 (2017).
- [SpencerJWolfSLKesarTM] Spencer, J., Wolf, S.L., Kesar, T.M. Biofeedback for post-stroke gait retraining: a review of current evidence and future research directions in the context of emerging technologies. *front. neurol.* 12, 637199 (2021).
- [SremakaewMJullGTreleavenJUthaikhupS] Sremakaew, M., Jull, G., Treleaven, J., Uthaikhup, S. Effectiveness of adding rehabilitation of cervical related sensorimotor control to manual therapy and exercise for neck pain: a randomized controlled trial. *musculoskelet. sci. pract.* 63, 102690 (2023). <https://doi.org/10.1016/j.msksp.2022.102690>.
- [SulisEMarianiSMontagnaS] Sulis, E., Mariani, S., Montagna, S. A survey on agents applications in healthcare: opportunities, challenges and trends. *comput. methods programs biomed.* 107525 (2023).
- [TreleavenJa] Treleaven, J. Dizziness, unsteadiness, visual disturbances, and sensorimotor control in traumatic neck pain. *j. orthop. sports phys. ther.* 47 (7), 492–502 (2017). <https://doi.org/10.2519/jospt.2017.7052>.
- [TreleavenJb] Treleaven, J. Sensorimotor disturbances in neck disorders affecting postural stability, head and eye movement control. *man. ther.* 13 (1), 2–11 (2008). <https://doi.org/10.1016/j.math.2007.06.003>.
- [VinoskiS] Vinoski, S. Advanced message queuing protocol. *ieee internet comput.* 10 (6), 87–89 (2006). <https://doi.org/10.1109/mic.2006.116>.
- [WeizmanYTiroshOFussFKTanAMRutzE] Weizman, Y., Tirosh, O., Fuss, F.K., Tan, A.M., Rutz, E. Recent state of wearable imu sensors use in people living with spasticity: a systematic review. *sensors* 22 (5), 1791 (2022).
- [YoonTLKimHNMinJH] Yoon, T.-L., Kim, H.-N., Min, J.-H. Validity and reliability of an inertial measurement unit-based 3-dimensional angular measurement of cervical range of motion. *j. manipul. physiol. ther.* 42 (1), 75–81 (2019).
- [ZoeteRMJOsmotherlyPGRivettDASnodgrassSJ] Zoete, R.M.J., Osmotherly, P.G., Rivett, D.A., Snodgrass, S.J. Seven cervical sensorimotor control tests measure different skills in individuals with chronic idiopathic neck pain. *braz. j. phys. ther.* 24 (1), 69–78 (2020). <https://doi.org/10.1016/j.bjpt.2018.10.013>.
- [nis25] NIST Chemistry WebBook, NIST Standard Reference Database Number 69. 2025. URL: <https://webbook.nist.gov/chemistry/> (visited on 2026-05-25), doi:10.18434/T4D303.
- [Ada09] Victor S. Adamchik. Algorithmic complexity. 2009. Carnegie Mellon University, 15-121 Introduction to Data Structures lecture notes. URL: <https://viterbi-web.usc.edu/TU\textbackslash{}textasciitildeadamchik/15-121/lectures/Algorithmic\TU\textbackslash{}%20Complexity/complexity.html> (visited on 2026-05-26).
- [ABucicS25] Noga Alon, Matija Bucić, and Lisa Sauermann. Unit and distinct distances in typical norms. *Geometric and Functional Analysis*, 35(1):1–42, 2025. doi:10.1007/s00039-025-00698-x.
- [Bra96] Peter Brass. Erdős distance problems in normed spaces. *Computational Geometry*, 6(4):195–214, 1996. doi:10.1016/0925-7721(95)00019-4.
- [BMP05] Peter Brass, William O. J. Moser, and János Pach. *Research Problems in Discrete Geometry*. Springer, New York, 2005. ISBN 9780387238159. doi:10.1007/0-387-29929-7.
- [BilkaBF+10] Ondřej Bílka, Kevin Buchin, Radoslav Fulek, Masashi Kiyomi, Yoshio Okamoto, Shin-ichi Tanigawa, and Csaba D. Tóth. A tight lower bound for convexly independent subsets of the minkowski sums of planar point sets. *Electronic Journal of Combinatorics*, 17(1):N35, 2010. doi:10.37236/484.

- [Dav00] Harold Davenport. *Multiplicative Number Theory*. Volume 74 of Graduate Texts in Mathematics. Springer, New York, third edition, 2000. ISBN 9780387950976. Revised by Hugh L. Montgomery. doi:10.1007/978-1-4757-5927-3.
- [Dir30] P. A. M. Dirac. A theory of electrons and protons. *Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character*, 126(801):360–365, 1930. doi:10.1098/rspa.1930.0013.
- [DdSMS99] John D. Dixon, Marcus P. F. du Sautoy, Avinoam Mann, and Dan Segal. *Analytic Pro- p Groups*. Volume 61 of Cambridge Studies in Advanced Mathematics. Cambridge University Press, Cambridge, second edition, 1999. ISBN 9780521650113.
- [Ein20] Albert Einstein. *Relativity: The Special and General Theory*. Henry Holt and Company, New York, 1920.
- [EPRothvossS08] Friedrich Eisenbrand, János Pach, Thomas Rothvoß, and Nir B. Sopher. Convexly independent subsets of the minkowski sum of planar point sets. *Electronic Journal of Combinatorics*, 15(1):N8, 2008. doi:10.37236/883.
- [ErdHos46] Paul Erdős. On sets of distances of n points. *American Mathematical Monthly*, 53(5):248–250, 1946. doi:10.2307/2305092.
- [ErdHosF97] Paul Erdős and Peter C. Fishburn. Minimum planar sets with maximum equidistance counts. *Computational Geometry*, 7(4):207–218, 1997. doi:10.1016/0925-7721(95)00050-X.
- [FRB16] Richard M. Felder, Ronald W. Rousseau, and Lisa G. Bullard. *Elementary Principles of Chemical Processes*. Wiley, Hoboken, NJ, fourth edition, 2016. ISBN 9781118431221.
- [GS64] E. S. Golod and I. R. Shafarevich. On the class field tower. *Izvestiya Akademii Nauk SSSR. Seriya Matematicheskaya*, 28(2):261–272, 1964. English translation: American Mathematical Society Translations, Series 2, 48 (1965), 91–102.
- [GS65] E. S. Golod and I. R. Shafarevich. On class field towers. In *Fourteen Papers on Logic, Algebra, Complex Variables and Topology*, volume 48 of American Mathematical Society Translations, Series 2, pages 91–102. American Mathematical Society, Providence, RI, 1965.
- [GST25] Josef Greilhuber, Carl Schildkraut, and Jonathan Tidor. More unit distances in arbitrary norms. *Bulletin of the London Mathematical Society*, 57(9):2885–2901, 2025. doi:10.1112/blms.70133.
- [GK15] Larry Guth and Nets Hawk Katz. On the erdős distinct distances problem in the plane. *Annals of Mathematics*, 181(1):155–190, 2015. doi:10.4007/annals.2015.181.1.2.
- [Haa83] S. E. Haaland. Simple and explicit formulas for the friction factor in turbulent pipe flow. *Journal of Fluids Engineering*, 105(1):89–90, 1983. doi:10.1115/1.3240948.
- [HM01] Farshid Hajir and Christian Maire. Asymptotically good towers of global fields. In Carles Casacuberta, Rosa Maria Miró-Roig, Joan Verdera, and Sebastià Xambó-Descamps, editors, *European Congress of Mathematics, Vol. II (Barcelona, 2000)*, volume 202 of Progress in Mathematics, pages 207–218. Birkhäuser, Basel, 2001. doi:10.1007/978-3-0348-8266-8_16.
- [HMR21] Farshid Hajir, Christian Maire, and Ravi Ramakrishna. Cutting towers of number fields. *Annales Mathématiques du Québec*, 45(2):321–345, 2021. doi:10.1007/s40316-021-00156-8.
- [Hod96] Derek Hodson. Laboratory work as scientific method: three decades of confusion and distortion. *Journal of Curriculum Studies*, 28(2):115–135, 1996. doi:10.1080/0022027980280201.
- [JG11] Kenneth A. Johnson and Roger S. Goody. The original michaelis constant: translation of the 1913 michaelis-menten paper. *Biochemistry*, 50(39):8264–8269, 2011. doi:10.1021/bi201284u.
- [Koc02] Helmut Koch. *Galois Theory of p -Extensions*. Springer Monographs in Mathematics. Springer, Berlin and Heidelberg, 2002. ISBN 9783540436294. doi:10.1007/978-3-662-04967-9.
- [KHovariSosTuran54] Tamás Kővári, Vera T. Sós, and Pál Turán. On a problem of k. zarankiewicz. *Colloquium Mathematicum*, 3:50–57, 1954. doi:10.4064/cm-3-1-50-57.
- [Lan94] Serge Lang. *Algebraic Number Theory*. Volume 110 of Graduate Texts in Mathematics. Springer, New York, second edition, 1994. ISBN 9780387942254. doi:10.1007/978-1-4612-0853-2.
- [Matouvsek11] Jiří Matoušek. The number of unit distances is almost linear for most norms. *Advances in Mathematics*, 226(3):2618–2628, 2011. doi:10.1016/j.aim.2010.09.009.

- [MT25] Warren L. McCabe and Ernest W. Thiele. Graphical design of fractionating columns. *Industrial \& Engineering Chemistry*, 17(6):605–611, 1925. doi:10.1021/ie50186a023.
- [MBR+24] André Filipe Sales Mendes, Héctor Sánchez San Blas, Fátima Pérez Robledo, Juan F. De Paz Santana, and Gabriel Villarrubia González. A novel multiagent system for cervical motor control evaluation and individualized therapy: integrating gamification and portable solutions. *Multimedia Systems*, 2024. doi:10.1007/s00530-024-01328-6.
- [MM13] Leonor Michaelis and Maud Leonora Menten. Die kinetik der invertinwirkung. *Biochemische Zeitschrift*, 49:333–369, 1913.
- [Moo44] Lewis F. Moody. Friction factors for pipe flow. *Transactions of the ASME*, 66(8):671–678, 1944. doi:10.1115/1.4018140.
- [Mul18] Julio Mulero. La música de las figuras de lissajous. 2018. El blog de Dimates, Universidad de Alicante. URL: <https://blogs.ua.es/dimates/2018/09/25/la-musica-de-las-figuras-de-lissajous/> (visited on 2026-05-26).
- [Neu99] Jürgen Neukirch. *Algebraic Number Theory*. Volume 322 of Grundlehren der mathematischen Wissenschaften. Springer, Berlin and Heidelberg, 1999. ISBN 9783540653998. Translated from the German by Norbert Schappacher; with a foreword by G. Harder. doi:10.1007/978-3-662-03983-0.
- [NSW08] Jürgen Neukirch, Alexander Schmidt, and Kay Wingberg. *Cohomology of Number Fields*. Volume 323 of Grundlehren der mathematischen Wissenschaften. Springer, Berlin and Heidelberg, second edition, 2008. ISBN 9783540378884. doi:10.1007/978-3-540-37889-1.
- [RZ10] Luis Ribes and Pavel Zalesskii. *Profinite Groups*. Volume 40 of Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge. Springer, Berlin and Heidelberg, second edition, 2010. ISBN 9783642016417. doi:10.1007/978-3-642-01642-4.
- [Sha63] Igor R. Shafarevich. Extensions à points de ramification donnés. *Publications Mathématiques de l’IHÉS*, 18:71–92, 1963. In Russian. doi:10.1007/BF02684785.
- [Sha66] Igor R. Shafarevich. Extensions with given points of ramification. *American Mathematical Society Translations, Series 2*, 59:128–149, 1966. English translation by J. W. S. Cassels; also cited as Extensions with prescribed ramification points.
- [SSzemerédiT84] Joel H. Spencer, Endre Szemerédi, and William T. Trotter. Unit distances in the euclidean plane. In Béla Bollobás, editor, *Graph Theory and Combinatorics*, pages 293–303. Academic Press, London, 1984.
- [Szekely97] László A. Székely. Crossing numbers and hard erdős problems in discrete geometry. *Combinatorics, Probability and Computing*, 6(3):353–358, 1997. doi:10.1017/S0963548397002976.
- [Thi39] Ernest W. Thiele. Relation between catalytic activity and size of particle. *Industrial \& Engineering Chemistry*, 31(7):916–920, 1939. doi:10.1021/ie50355a027.
- [TS13] Gavin Towler and Ray Sinnott. *Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design*. Butterworth-Heinemann, Oxford, second edition, 2013. ISBN 9780080966595.
- [Tsc26] N. Tschebotareff. Die bestimmung der dichtigkeit einer menge von primzahlen, welche zu einer gegebenen substitutionsklasse gehören. *Mathematische Annalen*, 95(1):191–228, 1926. doi:10.1007/BF01206606.
- [VGO+20] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, and others. Scipy 1.0: fundamental algorithms for scientific computing in python. *Nature Methods*, 17(3):261–272, 2020. doi:10.1038/s41592-019-0686-2.
- [Was97] Lawrence C. Washington. *Introduction to Cyclotomic Fields*. Volume 83 of Graduate Texts in Mathematics. Springer, New York, second edition, 1997. ISBN 9781461273462. doi:10.1007/978-1-4612-1934-7.
- [WP54] P. B. Weisz and C. D. Prater. Interpretation of measurements in experimental catalysis. *Advances in Catalysis*, 6:143–196, 1954. doi:10.1016/S0360-0564(08)60390-9.
- [Zim00] John M. Ziman. *Real Science: What It Is, and What It Means*. Cambridge University Press, Cambridge, 2000. ISBN 9780521772297.
- [AgostonPalvolgyi22] Péter Ágoston and Dömötör Pálvolgyi. An improved constant factor for the unit distance problem. *Studia Scientiarum Mathematicarum Hungarica*, 59(1):40–57, 2022. doi:10.1556/012.2022.01517.

- [AbdollahiMKuberPMSHiraishiMSoangraRRashediE] Abdollahi, M., Kuber, P.M., Shiraishi, M., Soangra, R., Rashedi, E. Kinematic analysis of 360 turning in stroke survivors using wearable motion sensors. *sensors* 22 (1), 385 (2022).
- [AiQLiuZMengWLiuxQXieSQ] Ai, Q., Liu, Z., Meng, W., Liu, Q., Xie, S.Q. Machine learning in robot assisted upper limb rehabilitation: a focused review. *ieee transactions on cognitive and developmental systems* (2021).
- [BarraOrtizHAMMatamalaAMInostrozaFLArayaCLMondacaVN] Barra Ortiz, H.A., Matamala, A.M., Inostroza, F.L., Araya, C.L., Mondaca, V.N. Efficacy of biofeedback in rehabilitation of musculoskeletal disorders: a systematic review. *postepy rehabilitacji* 36 (1), 41 (2022).
- [BarzegarKhanghahAFernieGRFekrA] Barzegar Khanghah, A., Fernie, G., Roshan Fekr, A. Design and validation of vision-based exercise biofeedback for tele-rehabilitation. *sensors* 23 (3), 1206 (2023).
- [BisioIGaribottoCLavagettoFSciarroneA] Bisio, I., Garibotto, C., Lavagetto, F., Sciarrone, A. When ehealth meets iot: a smart wireless system for post-stroke home rehabilitation. *ieee wirel. commun.* 26 (6), 24–29 (2019) subjects with whiplash-associated disorders or non-specific neck pain and healthy controls. *clin. biomech.* 22 (8), 865–873 (2007). <https://doi.org/10.1016/j.clinbiomech.2007.05.008>.
- [BlasHSSMendesASEncinasFGSilvaLAGonzalezGV] Blas, H.S.S., Mendes, A.S., Encinas, F.G., Silva, L.A., Gonzalez, G.V. A multiagent system for data fusion techniques applied to the internet of things enabling physical rehabilitation monitoring. *appl. sci.* 11 (1), 331 (2020).
- [CalvaresiDMarinoniMDragoniAFHilfikerRSchumacherM] Calvaresi, D., Marinoni, M., Dragoni, A.F., Hilfiker, R., Schumacher, M. Realtime multi-agent systems for telerehabilitation scenarios. *artif. intell. med.* 96, 217–231 (2019).
- [ChenJorCKChenT] Chen, J., Or, C.K., Chen, T. Effectiveness of using virtual reality– supported exercise therapy for upper extremity motor rehabilitation in patients with stroke: systematic review and meta-analysis of randomized controlled trials. *j. med. internet res.* 24 (6), 24111 (2022).
- [ChenKBSestoMEPontoKLeonardJMasonAVanderheidenGWilliamsJRadwinRG] Chen, K.B., Sesto, M.E., Ponto, K., Leonard, J., Mason, A., Vanderheiden, G., Williams, J., Radwin, R.G. Use of virtual reality feedback for patients with chronic neck pain and kinesiophobia. *ieee trans. neural syst. rehabil. eng.* 25 (8), 1240–1248 (2016).
- [DePauwJMercelisRHallemansAMichielsSTruijenSCrasPDHertoghW] De Pauw, J., Mercelis, R., Hallemans, A., Michiels, S., Truijen, S., Cras, P., De Hertogh, W. Cervical sensorimotor control in idiopathic cervical dystonia: a cross-sectional study. *brain behav.* 7 (9), 00735 (2017). <https://doi.org/10.1002/brb3.735>.
- [DonSDKooningMVoogtLickmansKDaenenLNijsJ] Don, S., De Kooning, M., Voogt, L., Ickmans, K., Daenen, L., Nijs, J. The effect of visual feedback of the neck during movement in people with chronic whiplashassociated disorders: an experimental study. *j. orthop. sports phys. ther.* 47 (3), 190–199 (2017). <https://doi.org/10.2519/jospt.2017.689110.2519/jospt.2017.6891131> page 18 of 18 a. f. sales mendes et al.
- [DuQBaiHZhuZ] Du, Q., Bai, H., Zhu, Z. Intelligent evaluation method of human cervical vertebra rehabilitation based on computer vision. *sensors* 23 (8), 3825 (2023).
- [fstfirenze15] fstfirenze. Lissajous's figures, Wheatstone apparatus for compounding two orthogonal oscillations. 2015. YouTube video, published 2015-08-17. URL: <https://www.youtube.com/watch?v=xPjNPb8h8Ok> (visited on 2026-05-26).
- [GrazioliETranchitaEBorrielloGCerulliCMingantiCParisiA] Grazioli, E., Tranchita, E., Borriello, G., Cerulli, C., Minganti, C., Parisi, A. The effects of concurrent resistance and aerobic exercise training on functional status in patients with multiple sclerosis. *curr. sports med. reports* 18, 452–457 (2019). <https://doi.org/10.1249/jsr.0000000000000661>.
- [GuidanceM] Guidance, M. Motion guidance. [https:// www. motio nguid ance. com/](https://www.motio nguid ance. com/). accessed 31 agu 2023.
- [HidalgoPerezAFGarciaALdUraldeVillanuevaIGMartinezAPAlemanyaAFernandezCarneroJLToucheR] Hidalgo-Peréz, A., Fernández-García, Á., López-de-UraldeVillanueva, I., Gil-Martínez, A., Paris-Aleman, A., FernándezCarnero, J., La Touche, R. Effectiveness of a motor control therapeutic exercise program combined with motor imagery on the sensorimotor function of the cervical spine: a randomized controlled trial. *int. j. sports phys. ther.* 10 (6), 877–892 (2015).

- [HumayunMJhanjhiNZAlmotilagAAlmufarehMF] Humayun, M., Jhanjhi, N.Z., Almotilag, A., Almufareh, M.F. Agent-based medical health monitoring system. *sensors* 22 (8), 2820 (2022).
- [InternationalUoPaACchemistry22] International Union of Pure and Applied Chemistry. Periodic table of elements. 2022. Latest release dated 2022-05-04. URL: <https://iupac.org/what-we-do/periodic-table-of-elements/> (visited on 2026-05-25).
- [InternationalUoPaACchemistry25a] International Union of Pure and Applied Chemistry. Henderson–hasselbalch equation. IUPAC Compendium of Chemical Terminology, 5th ed., 2025. Online version 5.0.0. URL: <https://goldbook.iupac.org/terms/view/H02781> (visited on 2026-05-25), doi:10.1351/goldbook.H02781.
- [InternationalUoPaACchemistry25b] International Union of Pure and Applied Chemistry. Nernst equation. IUPAC Compendium of Chemical Terminology, 5th ed., 2025. Online version 5.0.0. URL: <https://goldbook.iupac.org/terms/view/09068> (visited on 2026-05-25), doi:10.1351/goldbook.09068.
- [KazeminasabSNejadghaderiSAAmiriPPourfathiHAKhodaeiMSullmanMJKolahiAASafiriS] Kazeminasab, S., Nejadghaderi, S.A., Amiri, P., Pourfathi, H., Araj-Khodaei, M., Sullman, M.J., Kolahi, A.-A., Safiri, S. Neck pain: global epidemiology, trends and risk factors. *bmc musculoskelet. disord.* 23 (1), 1–13 (2022).
- [KheraPKumarN] Khera, P., Kumar, N. Role of machine learning in gait analysis: a review. *j. med. eng. technol.* 44 (8), 441–467 (2020).
- [KimWSChoSKuJKimYLeeKHwangHJPaikNJ] Kim, W.-S., Cho, S., Ku, J., Kim, Y., Lee, K., Hwang, H.-J., Paik, N.-J. Clinical application of virtual reality for upper limb motor rehabilitation in stroke: review of technologies and clinical evidence. *j. clin. med.* 9 (10), 3369 (2020).
- [KumarSRavellaUKShrivastavaAKAnilMSwamySK] Kumar, S., Ravella, U.K., Shrivastava, A.K., Anil, M., Swamy, S.K. A review on human cervical fatigue measurement technologies and data analysis methods.
- [LaranjoLDunnAGTongHLKocaballiABChenJBashirRSurianDGallegoBMagrabiFLauAYeal] Laranjo, L., Dunn, A.G., Tong, H.L., Kocaballi, A.B., Chen, J., Bashir, R., Surian, D., Gallego, B., Magrabi, F., Lau, A.Y., et al. Conversational agents in healthcare: a systematic review. *j. am. med. inform. assoc.* 25 (9), 1248–1258 (2018).
- [MalflietAKregelJCoppietersIPauwRDMeeusMRousselNCagnieBDanneelsLNijsJ] Malfliet, A., Kregel, J., Coppieters, I., Pauw, R.D., Meeus, M., Roussel, N., Cagnie, B., Danneels, L., Nijs, J. Effect of pain neuroscience education combined with cognition-targeted motor control training on chronic spinal pain. *jama neurol.* 75 (7), 808 (2018). <https://doi.org/10.1001/jaman.neurol.2018.0492>.
- [MichielsSHertoghWDTuijtenSNovemberDWuytsFHeyningPV] Michiels, S., Hertogh, W.D., Tuijten, S., November, D., Wuyts, F., Heyning, P.V. The assessment of cervical sensory motor control: a systematic review focusing on measuring methods and their clinimetric characteristics. *gait posture* 38 (1), 1–7 (2013). <https://doi.org/10.1016/j.gaitp.ost.2012.10.007>.
- [MihajlovicZPopovicSBrkicKCosicK] Mihajlovic, Z., Popovic, S., Brkic, K., Cosic, K. A system for head-neck rehabilitation exercises based on serious gaming and virtual reality. *multimed. tools appl.* 77 , 19113–19137 (2018).
- [OpenAI26] OpenAI. Planar Point Sets with Many Unit Distances. 2026. Manuscript supplied as course material. URL: <https://cdn.openai.com/pdf/74c24085-19b0-4534-9c90-465b8e29ad73/unit-distance-proof.pdf> (visited on 2026-05-26).
- [ParkJSJungYJLeeG] Park, J.-S., Jung, Y.-J., Lee, G. Virtual reality-based cognitive– motor rehabilitation in older adults with mild cognitive impairment: a randomized controlled study on motivation and cognitive function. *healthcare* 8 , 335 (2020).
- [PavelValtr05] Pavel Valtr. Strictly convex norms allowing many unit distances and related touching questions. 2005. Manuscript, Charles University, Prague.
- [PengBYangLLiYLiutLiuYa] Peng, B., Yang, L., Li, Y., Liu, T., Liu, Y. Cervical proprioception impairment in neck pain-pathophysiology, clinical evaluation, and management: a narrative review. *pain ther.* 10 (1), 143–164 (2021). <https://doi.org/10.1007/s40122-020-00230-z>.
- [PengBYangLLiYLiutLiuYb] Peng, B., Yang, L., Li, Y., Liu, T., Liu, Y. Cervical proprioception impairment in neck pain-pathophysiology, clinical evaluation, and management: a narrative review. *pain ther.* 10 , 143–164 (2021).

- [PIEoscarUCM12] PIE “oscar” UCM. Figuras de Lissajous / Lissajous figures. 2012. YouTube video, published 2012-06-18. URL: <https://www.youtube.com/watch?v=mLhXPBvllWw> (visited on 2026-05-26).
- [ProjectJupyterBC+25] Project Jupyter, Evan Bolyen, J. Gregory Caporaso, Rowan Cockett, Danny Garside, Chris Holdgraf, Angus Hollands, Julia Kent, Franklin Koch, Jim Madge, Matthew McKay, Mike Morrison, Fernando Pérez, Steve Purves, Min Ragan Kelley, Brian E. J. Rose, Malvika Sharan, Brigitta M. Sipőcz, Stéfan J. Walt, and Kirstie Jane Whitaker. Jupyter Book 2 and the MyST Document Stack: A Modular, Extensible, Web-Native Stack for Authoring and Publishing Computational Narratives. In *Proceedings of the 24th Python in Science Conference*, 173–193. 2025. URL: <https://doi.org/10.25080/hwcj9957>, doi:10.25080/hwcj9957.
- [QuNTianHDMartinoEZhangB] Qu, N., Tian, H., De Martino, E., Zhang, B. Neck pain: do we know enough about the sensorimotor control system? *front. comput. neurosci.* (2022). <https://doi.org/10.3389/fncom.2022.946514>.
- [QuNTianHDMartinoEDZhangB] Qu, N., Tian, H., Martino, E.D., Zhang, B. Neck pain: do we know enough about the sensorimotor control system? *front. comput. neurosci.* (2022). <https://doi.org/10.3389/fncom.2022.946514>.
- [RiffittsMOhAAlemuAPatelVSmithCNMuratiSBailesAAllenMDombrowskiMLeeJYea] Riffitts, M., Oh, A., Alemu, A., Patel, V., Smith, C.N., Murati, S., Bailes, A., Allen, M., Dombrowski, M., Lee, J.Y., et al. Functional range of motion of the cervical spine in cervical fusion patients during activities of daily living. *j. biomech.* 152, 111528 (2023).
- [RodaCRodriguezACLJaqueroVNavarroEGonzalezP] Roda, C., Rodríguez, A.C., López-Jaquero, V., Navarro, E., González, P. A multi-agent system for acquired brain injury rehabilitation in ambient intelligence environments. *neurocomputing* 231, 11–18 (2017).
- [SpencerJWolfSLKesarTM] Spencer, J., Wolf, S.L., Kesar, T.M. Biofeedback for post-stroke gait retraining: a review of current evidence and future research directions in the context of emerging technologies. *front. neurol.* 12, 637199 (2021).
- [SremakaewMJullGTreleavenJUthaikhupS] Sremakaew, M., Jull, G., Treleaven, J., Uthaikhup, S. Effectiveness of adding rehabilitation of cervical related sensorimotor control to manual therapy and exercise for neck pain: a randomized controlled trial. *musculoskelet. sci. pract.* 63, 102690 (2023). <https://doi.org/10.1016/j.msksp.2022.102690>.
- [SulisEMarianiSMontagnaS] Sulis, E., Mariani, S., Montagna, S. A survey on agents applications in healthcare: opportunities, challenges and trends. *comput. methods programs biomed.* 107525 (2023).
- [TeachBooksDTeamTeachBooksusers25] TeachBooks Development Team and TeachBooks users. TeachBooks Manual. 2025. CC BY 4.0; last updated 2025-11-04. URL: <https://teachbooks.io/manual/intro.html> (visited on 2026-05-22).
- [TreleavenJa] Treleaven, J. Dizziness, unsteadiness, visual disturbances, and sensorimotor control in traumatic neck pain. *j. orthop. sports phys. ther.* 47 (7), 492–502 (2017). <https://doi.org/10.2519/jospt.2017.7052>.
- [TreleavenJb] Treleaven, J. Sensorimotor disturbances in neck disorders affecting postural stability, head and eye movement control. *man. ther.* 13 (1), 2–11 (2008). <https://doi.org/10.1016/j.math.2007.06.003>.
- [VinoskiS] Vinoski, S. Advanced message queuing protocol. *ieee internet comput.* 10 (6), 87–89 (2006). <https://doi.org/10.1109/mic.2006.116>.
- [WeizmanYTiroshOFussFKTanAMRutzE] Weizman, Y., Tirosh, O., Fuss, F.K., Tan, A.M., Rutz, E. Recent state of wearable imu sensors use in people living with spasticity: a systematic review. *sensors* 22 (5), 1791 (2022).
- [YoonTLKimHNMinJH] Yoon, T.-L., Kim, H.-N., Min, J.-H. Validity and reliability of an inertial measurement unit-based 3-dimensional angular measurement of cervical range of motion. *j. manipul. physiol. ther.* 42 (1), 75–81 (2019).
- [ZoeteRMJOsmotherlyPGRivettDASnodgrassSJ] Zoete, R.M.J., Osmotherly, P.G., Rivett, D.A., Snodgrass, S.J. Seven cervical sensorimotor control tests measure different skills in individuals with chronic idiopathic neck pain. *braz. j. phys. ther.* 24 (1), 69–78 (2020). <https://doi.org/10.1016/j.bjpt.2018.10.013>.